



A transformer-based neural ODE for dense prediction

Seyedalireza Khoshsirat¹ · Chandra Kambhamettu¹

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Abstract

Neural ordinary differential equations (ODEs) represent an emergent class of deep learning models exhibiting continuous depth. While they have shown promising results across various machine learning tasks, existing methods for dense prediction tasks have not fully harnessed their potential, often due to employing sub-optimal architectural components and limited dataset studies. To address this, our paper introduces a robust neural ODE architecture specifically tailored for dense prediction tasks and performs an extensive evaluation across a broad range of datasets. Our approach draws upon proven design elements from top-performing networks, integrating transformer blocks as core building blocks. Unique to our design is the retention of multiple concurrent representations at varying resolutions throughout the network. These representations continually exchange information, ensuring they remain updated. Our network achieves unrivaled performance in tasks such as image classification, semantic segmentation, and answer grounding. We conduct several ablation studies to shed light on the impacts of various design parameters. Our results affirm the effectiveness of our approach and its potential for further advancements in dense prediction tasks.

Keywords Neural ODE · Dense prediction · Transformer

1 Introduction

Neural ordinary differential equations (ODEs) have recently garnered attention for their potential in image classification [1] and semantic segmentation [2]. However, the existing methods for dense prediction [3–6] have been narrowly focused on semantic segmentation employing small, task-specific datasets. This specialization to smaller tasks hampers the models' capacity for achieving optimal accuracy and generalization on larger datasets. Addressing this limitation requires an expansive exploration of neural ODEs for dense prediction, encompassing larger datasets and diversifying beyond semantic segmentation.

In this paper, we introduce an innovative transformer-based neural ODE architecture that serves as a potent backbone for various computer vision tasks requiring dense predictions. In contrast to existing methods, we design a neural ODE inspired by renowned and top-performing methods. Our method is evaluated across several widely used vision datasets and establishes new benchmarks in diverse tasks.

For a fair comparison, we train and test existing neural ODE methods on large-scale datasets as well. Finally, we conduct exhaustive ablation studies to empirically validate the impact of our design decisions.

Subsequently, this section will provide an essential overview of neural ODEs and transformers.

1.1 Neural ordinary differential equations

In machine learning, we try to iteratively find a function that best describes the data. There are two basic approaches to finding this function. The first approach is to directly approximate the function by an analytical or numerical method. An ordinary linear regression falls into this category. The second approach is to approximate the derivative of the function. This results in an ordinary differential equation (ODE) which by solving, we get the approximation of the function. We can parameterize the derivative of the function as a neural network.

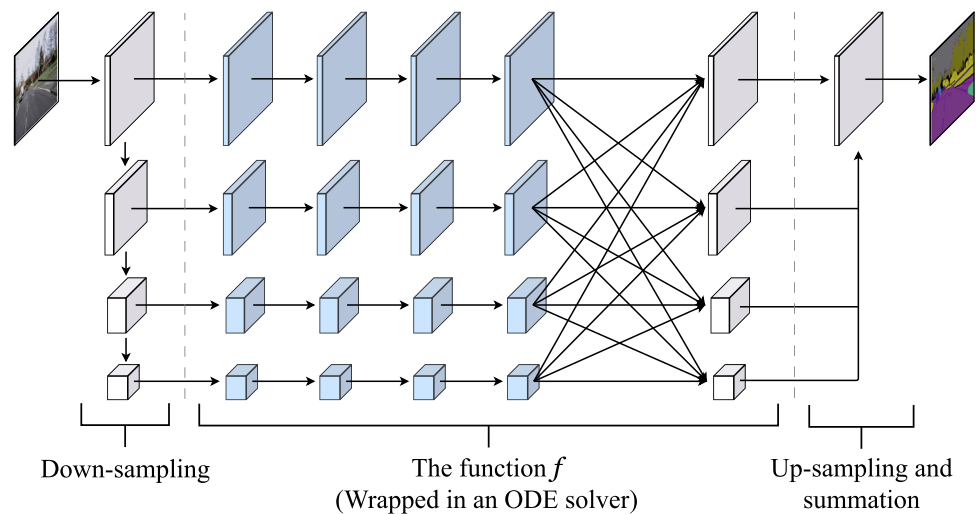
Consider a residual network [8] where all the hidden states have the same dimension. Such networks generate an output by doing a sequence of transformations to a hidden state [1]:

$$\mathbf{h}_{t+1} = \mathbf{h}_t + f(\mathbf{h}_t, \theta_t), \quad (1)$$

✉ Seyedalireza Khoshsirat
alireza@udel.edu

¹ Department of Computer and Information Sciences,
University of Delaware, Newark, DE 19716, USA

Fig. 1 The schematic layout of the proposed method. Each block shaded in blue signifies a Swin transformer block, as depicted in Fig. 2. In contrast, each block rendered in white denotes a patch merging block as detailed in [7]



where t is the layer depth and f is a neural layer. By adding an infinite number of layers, we get the continuous dynamics of hidden units using an ODE defined by a neural network [1]:

$$\frac{d\mathbf{h}(t)}{dt} = f(\mathbf{h}(t), t, \theta_t), \quad (2)$$

where $\mathbf{h}(t)$ is the value of the hidden state evaluated for some t and $f(\mathbf{h}(t), t, \theta_t)$ is a neural network layer parameterized by θ_t at layer t . By solving the integral:

$$\mathbf{h}(t) = \mathbf{h}(t_0) + \int_{t_0}^T f(\mathbf{h}(t), t, \theta_t) dt, \quad (3)$$

we can get the output value of a hidden layer at some depth T .

Neural ODEs take a different approach to backpropagate the loss through the continuous hidden state dynamics. In a neural ODE, the parameters include not only θ_t , but also the evaluation times t_0 and t_1 . Any differentiable loss function of our choice can be defined on these parameters:

$$\mathcal{L}(t_0, t_1, \theta_t) = \mathcal{L}(\text{ODESolve}(\mathbf{h}(t_0), t_0, t_1, \theta, f)). \quad (4)$$

A reverse traversal of the hidden states is required to compute the gradient of the loss with respect to the time (depth). Chen et al. [1] proposed using the adjoint method [9] to compute this gradient.

1.2 Transformers

Transformers are a class of models that have fundamentally revolutionized the field of natural language processing (NLP) [10]. The key innovation of transformers lies in the self-attention mechanism, which enables the model to weigh

the importance of different words in a sentence for a given task. While transformers have been predominantly applied in the domain of NLP, their architectural advantages have spurred researchers to adapt them for computer vision tasks. Computer vision presents unique challenges due to the high-dimensionality and spatial structure of visual data, which is inherently different from textual data. However, certain characteristics of transformers, such as their global receptive field and the ability to model long-range dependencies, are particularly suited for these challenges.

Swin transformer [11] is a transformer-based deep learning model with state-of-the-art performance in vision tasks. It introduced two key concepts to address the issues faced by the previous vision transformer networks, hierarchical feature-maps, and shifted window attention.

Swin transformer introduces an additional component to the original attention blocks in [10]. Relative position bias is an additional parametric bias term to encode the geometric relationship in self-attention calculation:

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V, \quad (5)$$

where $B \in \mathbb{R}^{M^2 \times M^2}$ is the relative position bias term for each head; $Q, K, V \in \mathbb{R}^{M^2 \times d}$ are the *query*, *key*, and *value* matrices; d is the *query/key* dimension, and M^2 is the number of patches in a window [11, 12].

2 Related work

2.1 Neural ODEs for dense prediction

Several studies have explored the integration of neural ODEs into deep learning architectures for the purpose of semantic segmentation. For instance, in [3], the authors propose an

innovative adaptation of the U-Net architecture [13], which is widely acknowledged for its efficacy in semantic segmentation tasks [14, 15]. Specifically, they incorporate neural ODEs into the U-Net structure. This is achieved by replacing the traditionally used residual blocks in each down-sampling and up-sampling pathway with a neural ODE that encompasses a single convolutional block. Similarly, in [4], the authors present a variant of the U-Net, though their approach differs from that in [3]. Instead of replacing the residual blocks with neural ODEs, a neural ODE block is added at the end of each branch.

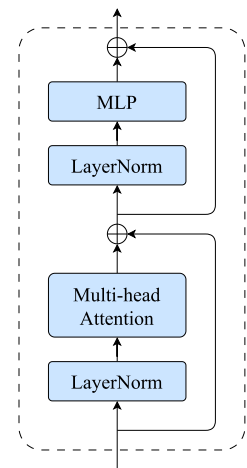
While U-Net has been instrumental in semantic segmentation, recent developments in architectural design have led to even more potent alternatives. The high-resolution network (HRNet) [7], for example, exhibits superior performance over U-Net, particularly in the realm of dense prediction tasks due to its ability to maintain high-resolution representations throughout the network. In light of these advancements, we propose a novel method that fuses the merits of HRNet with the expressive power of neural ODEs. By building on the HRNet framework, which has demonstrated higher accuracy than U-Net, and integrating neural ODEs, our method provides a significant boost in performance for dense prediction tasks.

In [2], the authors put forward an inventive architecture that employs neural ODEs, wherein residual blocks [8] serve as the fundamental building units. However, with the advent of attention mechanisms [10], there is growing evidence that transformer blocks, particularly when employed on extensive datasets, are outperforming residual blocks owing to their ability to capture long-range dependencies and contextual information more effectively [16]. In response to this development, our proposed architecture takes a departure from the conventional residual blocks and instead, harnesses the power of transformer blocks as its core building units.

On a different note, [17] brings to light an intriguing method that amalgamates neural ODEs with the level set method for image segmentation. In this hybrid approach, the derivative of the contour is parameterized using a neural ODE, which intrinsically learns a forcing function that governs the evolution of the contour over time. However, a notable limitation of this approach is its restriction to image segmentation tasks involving a single target class. This constraint can be rather limiting in real-world applications, where it is common to encounter images with multiple objects belonging to different classes. Our proposed method avoids this problem by providing a straightforward design inspired by the existing top-performing approaches.

At the time of this writing, neural ODEs are primarily employed in semantic segmentation and have not been utilized in other dense prediction tasks. Our research is the first to employ neural ODEs for other dense prediction tasks.

Fig. 2 The overview of a Swin transformer block [11]



2.2 Transformers

Unlike traditional transformers that process images as a flat array of pixels, methods such as Swin [11], CvT [18], Moat [19], VAN [20], and CSWin [21] enhance the local continuity of features and remove fixed-size position embedding to improve the performance of transformers in dense prediction tasks. Swin [11] hierarchically processes non-overlapping windows of an image and subsequently shifts these windows for more global representations. CvT [18] integrates convolutions into vision transformers to effectively capture local features while maintaining global context. Moat [19] alternates between mobile convolutions and attention mechanisms to create efficient and powerful vision models that achieve state-of-the-art performance on various visual recognition tasks. VAN [20] propose a novel linear attention named large kernel attention (LKA) to enable self-adaptive and long-range correlations in self-attention. CSWin [21] is an innovative vision transformer architecture that employs cross-shaped windows for self-attention, enhancing the model's capacity to capture both local and global features.

In Sect. 6.2, we conduct an ablation analysis to evaluate and contrast the performance of the aforementioned transformer designs when integrated into our proposed network.

2.3 Neural ODEs and transformers

A limited number of methods have attempted to integrate neural ODEs with transformers, specifically targeting image classification tasks. In [22], neural ODEs are used to formulate a variant of transformers that is depth-adaptive in the sense that the ODE solver takes an input-dependent number of time steps. Li et al. [23] showed that a transformer residual block can be described as a higher-order solution to ODEs. Based on this finding, they designed a new architecture for NLP tasks. In a similar work [24], a modified transformer layer based on a neural ODE interpretation is proposed that

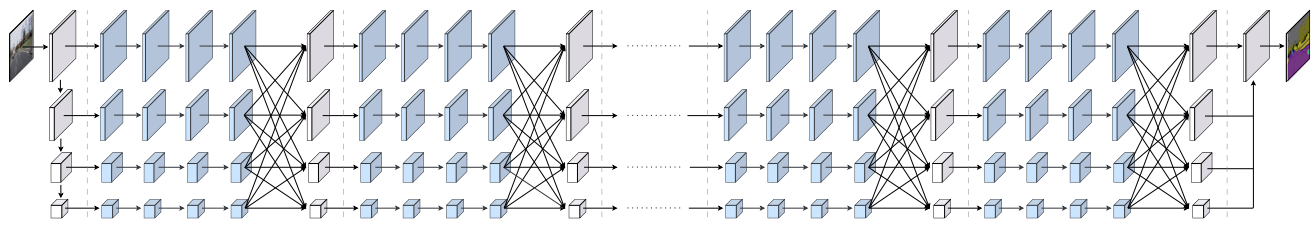


Fig. 3 The prototype network is created by repeating an HRNet module [7] with four branches at different feature-map resolutions. Each block shaded in blue signifies a Swin transformer block, as depicted

in Fig. 2, and each block rendered in white denotes a patch merging block as detailed in [7]. This network is not a neural ODE and is trained similarly to [11]

places the multi-head attention sub-layer and the MLP sub-layer parallel to each other. In [25], a lightweight architecture is proposed that couples continuous dynamical models (i.e., neural ODEs) with a proposed self-attention block to form a self-attention ODE solver.

As of the current state of research, there are no established methodologies that amalgamate neural ODEs with transformers for dense prediction tasks. Our contribution represents a pioneering effort in this domain.

3 Method

While existing neural ODE methods [2–4] excel in semantic segmentation on small, task-specific datasets, these techniques underperform when applied to large-scale datasets, as evidenced by the results in Tables 1, 2, and 3. Attempts to enhance their performance through model scaling and hyper-parameter fine-tuning were made.

In order to fully harness the potential of neural ODEs, we design a neural ODE architecture which demonstrates unprecedented accuracy across various computer vision tasks utilizing large-scale datasets. We start our design with a well-known and top-performing method. Based on this method, we design a prototype network that can be converted to a neural ODE. Next, we introduce a neural ODE that mirrors the prototype network. This phased approach allows for a comprehensive comparison of results at each stage: from a state-of-the-art network to the prototype network, and finally, to the neural ODE network.

3.1 Prototype network

One of the top-performing methods in dense prediction tasks is HRNet [7]. HRNet maintains high-resolution representations throughout the whole process. It starts from a high-resolution convolution stream, gradually adds high-to-low resolution convolution streams one by one, and connects the multi-resolution streams in parallel. The resulting network consists of several (four in the current design) stages,

and the n th stage contains n streams corresponding to n resolutions.

It is possible to convert a residual network into a neural ODE using the method presented in [1]. Therefore, we design our prototype network with a primary residual form using modules from HRNet. To this aim, we repeat a HRNet module in series multiple times and treat each instance as an individual residual unit, as visually represented in Fig. 3. Every module in this design features four branches, each presenting differing feature-map resolutions, and each branch contains multiple blocks. Throughout the experiments conducted for this paper, the principal module is reiterated six times to maintain a comparable number of parameters to HRNet.

The building blocks of HRNet are convolutional residual blocks [26]. To increase the learning capacity of our proposed network, we replace these convolutional blocks with Swin transformer blocks [11]. Figure 2 illustrates the overall design of a Swin transformer block. Similar to [11], we use patch merging to reduce the resolution by a factor of two for each stream.

3.2 TransNode

As the prototype network embodies an overall residual structure, it can be seamlessly transitioned into a neural ODE. In this form, either a single module or multiple modules collectively serve the role of the function f in Eq. 1. This module, or collection of modules, is encapsulated within an ODE solver. To generate the initial input feature-maps for the ODE solver, patch merging is employed in conjunction with an MLP, applied four times. The resulting four feature-maps are subsequently fed into the ODE solver. Notably, the output from the ODE solver maintains the same format as its input. By incorporating an additional four MLP layers, the channel count of the output feature-maps is adjusted to match the number of classes. Subsequently, these feature-maps are re-scaled to a higher resolution using bilinear interpolation and aggregated to yield the final output, as demonstrated in Fig. 1. Bilinear interpolation is specifically chosen to avoid

the emergence of the checkerboard artifact [28]. For the sake of brevity, this network is referred to as TransNode.

We use a compound loss function of RMI [30] and cross-entropy for the semantic segmentation and answer grounding tasks. Consequently, the overall loss function to be minimized is as follows:

$$\mathcal{L}_{\text{all}}(y, p) = \frac{1}{B} \sum_{b=1}^B \lambda \mathcal{L}_{\text{ce}}(y^{(b)}, p^{(b)}) + (1 - \lambda) \mathcal{L}_{\text{rmi}}(y^{(b)}, p^{(b)}) \quad (6)$$

where $\lambda \in [0, 1]$ is a weight factor, B denotes the number of samples in a mini-batch, $\mathcal{L}_{\text{ce}}(y^{(b)}, p^{(b)})$ is the standard cross-entropy loss between the b -th sample and its corresponding prediction, respectively, and $\mathcal{L}_{\text{rmi}}(y^{(b)}, p^{(b)})$ is the RMI loss as follows:

$$\mathcal{L}_{\text{rmi}}(\mathbf{Y}, \mathbf{P}) = \sum_{c=1}^C \frac{1}{2d} \text{Trace}(\log(\mathbf{M})) \quad (7)$$

and,

$$\mathbf{M} = \Sigma_{\mathbf{Y}} - \text{Cov}(\mathbf{Y}, \mathbf{P})(\Sigma_{\mathbf{P}}^{-1})^T \text{Cov}(\mathbf{Y}, \mathbf{P})^T \quad (8)$$

where $\mathbf{Y} = y^{(b)}$, $\mathbf{P} = p^{(b)}$, $\mathbf{M} \in \mathbb{R}^{d \times d}$, d denotes the number of pixels, C is the number of classes, $\Sigma_{\mathbf{Y}}$ is the variance matrix of $\mathbf{Y}$, and $\text{Cov}(\mathbf{Y}, \mathbf{P})$ is the covariance matrix of $\mathbf{Y}$ and $\mathbf{P}$. Similar to [30], we set $\lambda = 0.5$.

Regarding the ODE solver, we employ the DOPRI (Dormand-Prince-Shampine Runge-Kutta of order 5) provided by [1]. DOPRI [31] is an embedded method designed to solve ordinary differential equations. As a part of the Runge-Kutta family of ODE solvers, it specifically leverages six function evaluations to compute solutions with fourth- and fifth-order accuracy.

4 Experiments

We evaluate our approach in three vision tasks: image classification, semantic segmentation, and answer grounding.

4.1 Setup

Similar to HRNet [7], for both the prototype network and TransNode, we use a stem for the input image, which consists of two patch merging blocks to decrease the resolution to $1/4$, and is connected to the main body. The main body outputs the feature-maps with the same resolution ($1/4$), which are then made as large as the original resolution using bilinear interpolation. We use two modules in the main body to achieve the highest accuracy.

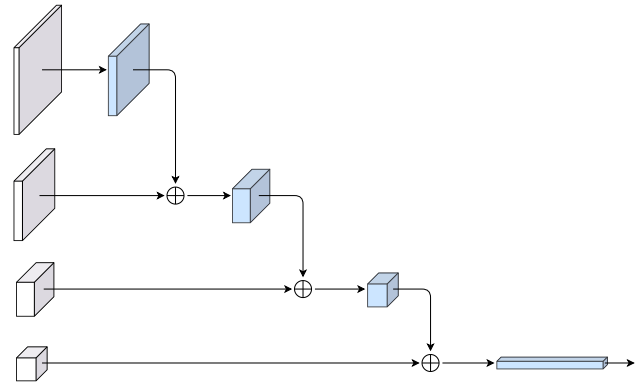


Fig. 4 This classifier head is attached to our neural ODE for image classification. At each step, the feature-maps are down-sampled, processed, and added to a lower-resolution one

4.2 Image classification

To show the learning capability of our neural ODE, we train and test it on a large-scale image classification dataset. We use ImageNet [27], which has 1.28M training images and 50k validation images in 1k classes, to evaluate our proposed method for image classification. We train the networks from scratch and report top-1 accuracy.

In order to adopt our proposed neural ODE for the image classification task, we augment our network with a classification head shown in Fig. 4. The classification head is described as follows. From higher to lower resolution, each feature-map goes through a patch merging block for down-sampling, and an MLP block for adaptation. Then, the resulting feature-map is added to its corresponding lower-resolution feature-map. This process is repeated for the resulting feature-map, until no lower-resolution feature-map is left. Finally, the last output is reshaped into a one-dimensional vector, and using an MLP block, the final logits vector is generated.

We use the same setup as in [11], i.e., we use AdamW optimizer [37] for 300 epochs and a weight decay of 0.05. The initial learning rate is 0.001 combined with a cosine learning rate scheduler with 20 epochs of linear warm-up.

4.3 Semantic segmentation

Semantic segmentation is the standard task for evaluating dense prediction methods. We evaluate our approach on four datasets: Cityscapes [38], CamVid [39], LIP [40], and PASCAL-Context [41]. We pre-train our prototype and neural ODE networks on ImageNet [27] and use the pre-trained networks in all our experiments. We use the mean Intersection over Union (mIoU) metric to compare all the methods. Similar to [11], for fine-tuning the pre-trained networks, we reduce the initial learning rate to 0.0001.

Cityscapes. The Cityscapes dataset [38] contains 5k pixel-level finely annotated street images and an additional

20k coarsely annotated images with 19 classes for evaluation. We train on the training, validation, and coarse sets to get the highest accuracy on the test set.

CamVid. The original version of the CamVid dataset [39] comprises 701 annotated images in 32 classes with a resolution of 960×720 . However, most literature only focuses on the protocol proposed in [42], which splits the dataset into 367 training, 101 validation, and 233 test images in 11 classes. We follow this protocol for training on CamVid.

LIP. The LIP dataset [40] contains 50,462 human images with detailed annotations. The model evaluation is done on 20 categories (including the background label). We follow the common testing protocol [7, 43] and resize the images to 473×473 .

PASCAL-Context. The PASCAL-Context dataset [41] adds annotations for more than 400 additional categories to the PASCAL VOC 2010 dataset. It contains 4,998 training

and 5,105 validation images. We follow [7, 29] and evaluate our method on 59 sub-categories.

4.4 Answer grounding

The answer grounding task is defined as detecting the pixels that can provide evidence for the answer to a given question regarding an image [44]. In other words, the task is to return the image regions used to arrive at the answer for a given visual question (question-image pair) with an answer. Although not every answer grounding datasets have dense ground truth labels, there are many that provide such labels. In this paper, we perform our experiments on four answer grounding datasets: TextVQA-X [32], VQS [33], VQA-X [45], and VizWiz-VQA-Grounding [44].

For each dataset, we use a top-performing method and replace its backbone network with our proposed TransNode. Additionally, since neural ODEs have not been evaluated on any of these datasets, we train and test SegNode [2] and compare it to TransNode.

TextVQA-X. The TextVQA-X dataset [32] (a subset of the TextVQA dataset [49] with manually annotated answer groundings) consists of 10,379 training images and 3,354 test images. Similar to the existing methods, we use the test set to evaluate our method and report the mean IoU measure.

VQS. The VQS dataset [33] builds upon the images, instance segmentation masks, and bounding boxes in COCO [50] and the questions and answers in the VQA dataset [51]. The VQS dataset contains 26,995 training, 5,000 validation, and 5,873 test images.

VQA-X. Since the VQA-X dataset [45] does not have annotations for its training set, we train our method on the training set of the VQS dataset [33]. The images of both datasets are from the COCO dataset [50]. The VQA-X dataset

Table 1 The top-1 and top-5 accuracy of image classification on ImageNet [27]

Method	Top-1	Top-5
ResNet-152 [26]	78.6	94.3
HRNet [7]	78.9	94.4
U-Node [3]	76.7	93.2
NODEs-UNet [4]	77.9	93.8
SegNode [2]	78.7	94.3
Prototype network	79.2	94.5
TransNode	79.4	94.8

Bold indicates the best performance among all the methods

The middle section lists the current neural ODE methods for dense prediction

Table 2 Comparison of results on four semantic segmentation datasets. The top section includes HRNet and its variants

Method	Cityscapes	CamVid	LIP	PASCAL-Context
HRNet [7]	81.6	80.9	55.9	54.0
HRNet+OCR [29]	83.0	81.7	56.6	56.2
HRNet+OCR [†] [29]	84.2	82.0	—	—
U-Node [3]	78.1	77.3	51.3	49.7
NODEs-UNet [4]	79.5	78.8	52.9	50.9
SegNode [2]	81.8	81.1	55.8	54.1
SegNode+OCR [2]	83.1	82.0	56.7	56.2
SegNode+OCR [†] [2]	84.5	82.2	—	—
Prototype network	82.1	81.2	56.1	54.4
TransNode	82.2	81.3	56.1	54.9
TransNode+OCR	83.5	82.2	57.1	56.8
TransNode+OCR [†]	84.8	82.6	—	—

Bold indicates the best performance among all the methods

The middle section lists existing neural ODE methods. The bottom section items are our proposed methods.

We use [†] to mark methods pre-trained on Mapillary

Table 3 The answer grounding results on two datasets, TextVQA-X [32] and VQS [33]

Method	TextVQA-X	VQS
MTXNet [32]	18.9	–
DeconvNet [33]	–	29.8
Mask Aggregation [33]	–	32.6
LXMERT [34]	24.9	33.3
VinVL [35]	25.6	33.9
MAC-Caps [36]	26.1	34.3
MAC-Caps + HRNet [7]	26.7	34.9
MAC-Caps + U-Node [3]	25.9	33.9
MAC-Caps + NODEs-UNet [4]	26.0	34.1
MAC-Caps + SegNode [2]	27.1	35.1
MAC-Caps + Prototype network	28.1	35.9
MAC-Caps + TransNode	28.6	36.3

Bold indicates the best performance among all the methods
We use MAC-Caps which achieves the best mIoU on both datasets

consists of 24,876 training, 1,431 validation, and 1,921 test images. We use the test set of the VQA-X dataset to evaluate our method and report the Rank Correlation as in [45].

VizWiz-VQA-Grounding. The VizWiz-VQA-Grounding dataset [44] contains 9,998 VQAs divided into 6,494/1,131/2,373 VQAs for training, validation, and testing. We use the VizWiz-VQA-Grounding Challenge online leaderboard [44] to evaluate the methods on the test set and report the mIoU measure.

5 Results

We report and analyze the results of each task separately in the following sections.

Table 4 Comparison of Rank Correlation on the test set of the VQA-X dataset [45]

Method	Rank Correlation
PJ-X [45]	0.342
CCM [46]	0.368
U-CAM [47]	0.372
Att-MFH [48]	0.376
Att-MCB [48]	0.396
Att-MCB + HRNet [7]	0.401
Att-MCB + U-Node [3]	0.388
Att-MCB + NODEs-UNet [4]	0.393
Att-MCB + SegNode [2]	0.404
Att-MCB + Prototype network	0.412
Att-MCB + TransNode	0.416

Bold indicates the best performance among all the methods
We use Att-MCB for this dataset since it achieves the best results

Table 5 The top-performing teams on the VizWiz-VQA-Grounding Challenge 2022 leaderboard [44]

Team	Mean IoU
MindX	66.9
MGTV	69.7
hsslab_inspur	70.1
GroundTruth	70.3
Aurora	70.6
Att-MCB + TransNode	71.5

Bold indicates the best performance among all the methods
At the time of this writing, none of the methods have been published

5.1 Image classification

Table 1 compares the top-1 and top-5 results of our method with HRNet [7] and current neural ODE methods for the image classification task on the ImageNet dataset [27]. Existing methods of neural ODE [3, 4] predominantly rely on a U-Net structure, which is made up of encoder and decoder components. The encoder segment primarily contributes to classification tasks, while the decoder is essentially beneficial for dense prediction tasks. As a result, in comparison with TransNode, these networks exhibit sub-optimal performance in image classification. The architecture of our proposed network is meticulously designed for efficiency, rendering it beneficial for both image classification and dense prediction tasks. The adaptability of our design allows it to excel across a variety of tasks, setting it apart from traditional neural ODE methods.

5.2 Semantic segmentation

Table 2 reports the performance of our method on four semantic segmentation datasets. Existing neural ODE methods [3, 4] gradually reduce the resolution of intermediate feature-maps to capture more semantic, high-level features. In contrast, TransNode maintains high-resolution representations throughout the model, preserving fine-grained spatial details that are critical for dense prediction tasks like semantic segmentation. Hence, our proposed method outperforms all existing methods, either when dealing with expansive datasets such as LIP, or high-resolution datasets, exemplified by Cityscapes.

5.3 Answer grounding

TransNode's design makes it flexible and adaptable to different computer vision tasks like answer grounding. Table 3 provides the comparison with several representative methods on the TextVQA-X [32] and VQS [33] datasets in terms of mIoU. Since MAC-Caps [36] is the best-performing method

Table 6 A comparison of a few important empirical computational measures on an NVIDIA Tesla V100 32GB for CamVid [39] dataset

Method	Maximum Batch Size for 32GB	Memory Usage for Batch Size 24 (GB)	Memory Usage for Testing One Image (GB)	Training Time per Epoch (Seconds)	Testing Time per Image (Milliseconds)	Number of Parameters (Millions)
U-Net [13]	36	21.8	0.8	10	4	31.0
PSPNet [52]	24	31.2	0.8	18	5	23.7
Deeplab v3 [53]	24	31.3	1.0	24	16	58.6
HRNet [7]	24	31.1	1.2	18	48	65.8
SegNode	62	13.4	0.7	34	117	20.9
Baseline network	26	30.9	1.2	19	49	70.9
TransNode	64	12.4	0.6	31	103	25.8

Bold indicates the best performance among all the methods

The training time per epoch is calculated using the maximum batch size possible. Our method requires the least amount of memory, but the longest computation time for training and testing

Table 7 The performance of five different transformer blocks integrated into our proposed network

Method	mIoU
VAN [20]	80.9
Moat [19]	81.2
CvT [18]	81.5
CSWin [21]	82.0
Swin [11]	82.2

on both datasets, we use MAC-Caps to evaluate the backbone networks for the answer grounding task.

We employ the best-performing method on the VQA-X dataset [45], which is Att-MCB [48], to compare the methods on this dataset. Table 4 reports the Rank Correlation measure for different methods.

Furthermore, we evaluate our method on the 2022 VizWiz-VQA-Grounding Challenge leaderboard [44]. Table 5 shows the current teams on the leaderboard.

TransNode simultaneously processes features at multiple resolutions, and allows for repeated multi-scale fusion of features across different resolutions. This multi-resolution, cross-scale interaction enhances the richness and completeness of the feature representation. Therefore, TransNode reaches a new state-of-the-art accuracy on each answer grounding dataset in our study.

6 Ablation STUDies

An ablation study examines the performance of a neural network by removing specific parts to understand the contribution of the part to the whole network [54]. In this section, we perform two ablation studies.

6.1 Empirical computational cost

This section provides an empirical comparison between our approach and a few well-known networks. We use an NVIDIA Tesla V100 32GB with the same network hyper-parameters as used before. All the experiments are implemented in Python using PyTorch. The results are calculated on CamVid dataset [39] with an image size of 480×360 .

Table 6 compares a few important empirical computational measures. Our method requires the least amount of memory, but the longest computation time for training and testing by a large margin. In particular, compared to HRNet [7], while our method has 61% less number of parameters, it requires 60% less memory for training and 50% less memory for testing. Conversely, HRNet requires 45% less training time and 54% less testing time.

6.2 Transformer blocks

Numerous transformer block designs have recently been proposed, each aiming to support multi-scale network architectures. This section provides a comparative performance analysis of five such transformer blocks when integrated into our proposed network. For the purpose of this comparison, each transformer block is assigned as the primary block in the network and subjected to a training and testing regimen on the Cityscapes dataset.

Table 7 encapsulates the outcomes of this investigation. The family of Swin transformers proves to be the most effective within the confines of these experiments. Given that Swin accomplishes the apex of accuracy, it is selected as the primary block for all experiments in this paper.

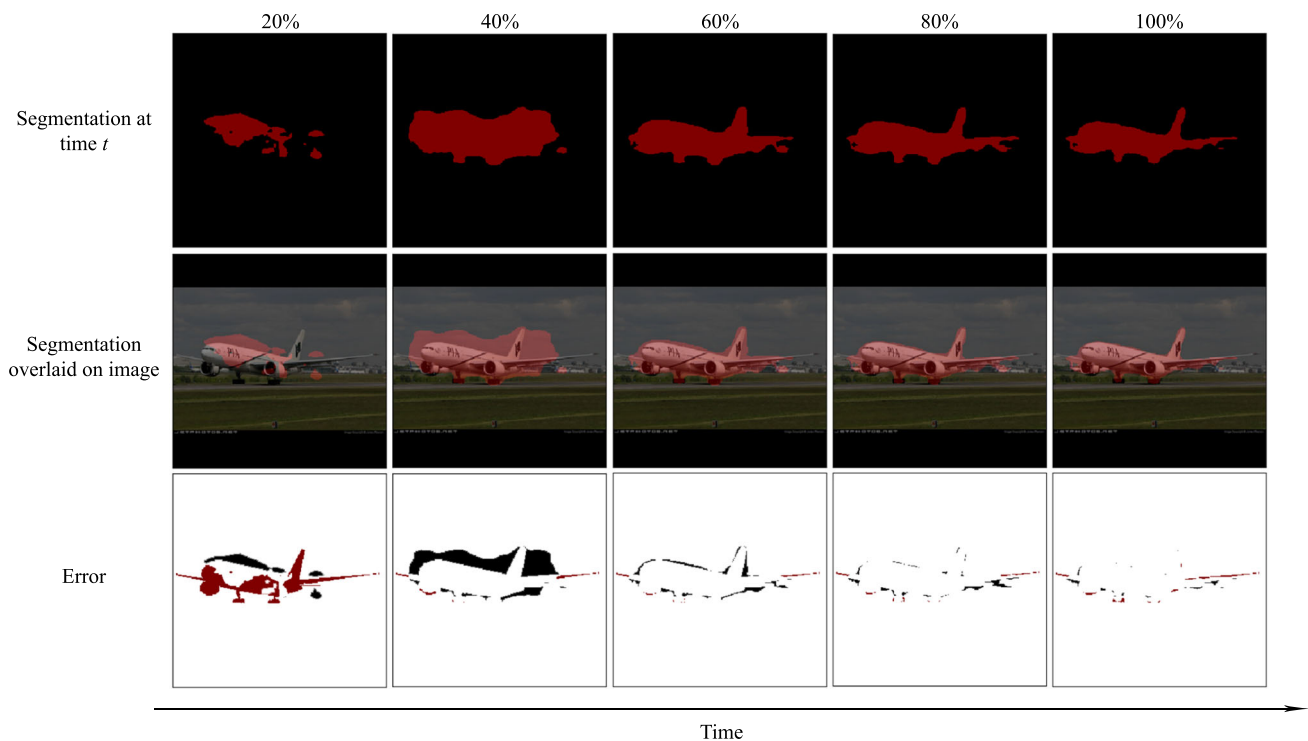


Fig. 5 Segmentation results from trajectories at different times. This image shows how the gradual transformations correct the segmentation over time

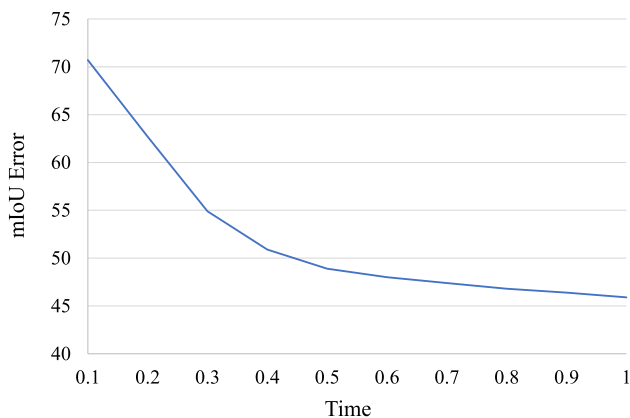


Fig. 6 The average mean IoU error of the trajectories during solving time, calculated on the PASCAL-Context validation set

6.3 Trajectories

Neural ODEs have the ability to trade-off speed and precision during both training and testing time. Most adaptive ODE solvers require the user to specify an error tolerance (both relative and absolute) that the solver will try to meet. The error tolerance is a way to trade-off compute time against the precision of the prediction. This is a more flexible approach in some ways than weight pruning or quantization, since tolerances can be adjusted throughout training, or even after the model has been deployed.

In this section, we analyze the speed-precision trade-off and show how the ODE solver gradually improves its output. Figure 5 visualizes the segmentation output of the network trajectories over time for one sample image. This figure shows the steps that the ODE solver takes during solving the network. To generate each step, the corresponding hyper-parameter of the ODE solver is modified to partially solve its input.

Figure 6 shows the average mean IoU error of the trajectories over time for all the samples in the PASCAL-Context validation set. One of the biggest issues with neural ODEs is that they require more computational resources during test time. To alleviate this problem, it is possible to sacrifice accuracy for speed. As an example, by sacrificing 3% of accuracy, the required computational time decreases by 50%.

7 Conclusion

Neural ODEs represent a novel class of deep learning model alternatives boasting implicit layers. Despite promising outcomes in image classification, existing methods for dense prediction often fall short, frequently due to using sub-optimal architectural components and employing smaller datasets. In this paper, we have proposed a neural ODE architecture specifically crafted for dense prediction tasks to address this problem. Our approach draws on the best prac-

tices from top-performing network designs across various tasks. Tested on a multitude of computer vision datasets, our proposed method delivered superior results in every experiment. To delve into the specifics of our method, we conducted numerous ablation studies.

Author Contributions Not applicable

Availability of data and materials Data available on request from the authors.

Code Availability Code available on request from the authors.

Declarations

Conflict of interest The authors did not receive support from any organization for the submitted work.

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Consent for publication Not applicable

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